A-level

ENVIRONMENTAL SCIENCE

Questions by topic: How humans influence biodiversity - Set 1

Mark scheme

AQA spec 3.1.2.2; The living environment; textbook pp. 20-25

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1 Food:** cod, tuna, swordfish, sharks, turtles – extinct - dodo, great auk, passenger pigeon **1 Fashion:** leopard, snow leopard, tiger, ocelot fur – fur seals – crocodile / alligator leather – kingfisher, parrot, ostrich feathers **1 Entertainment / pets:** parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants **1 Furniture / ornaments:** mahogany, teak, ramin timber – ebony and ivory piano keys – turtle shell, shark teeth, coral and shells as souvenirs **1 Traditional medicines:** tiger (bones, claws), rhinoceros horn, seahorses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	5

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Threat to humans:** sharks, venomous snakes, crocodiles **1 Disease vectors:** malaria mosquitoes, tsetse flies **1 Predators of livestock:** wolves, puma, birds of prey, herons, seals **1** Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** High demand and prices make illegal trade / poaching profitable – laws are hard to enforce across borders **1** Belief in the medicine persists despite little scientific evidence – species become rarer and more valuable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands – HEP schemes alter river levels flooding sandbank nests of turtles / birds – species with narrow tolerance (specialists) affected first – sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade **2** Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae **2** Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **abstraction** → water taken out of a river, lake or aquifer • **HEP** → electricity generated from falling water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6

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Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Reduces light penetration so submerged plants cannot photosynthesise **1** Blocks the gills of filter feeders such as bivalve molluscs – sediment smothers the bed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Rare species often have very specialised habitat requirements / narrow ranges of tolerance **1** A small change moves conditions outside their range while generalists survive – small populations cannot recover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) **1** As otters recovered after 1990 urchins declined and kelp recovered **1** Otter and kelp trends are positively correlated – urchins negatively correlated with both **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise **1** Sea urchins graze kelp so the kelp forest was over-grazed and destroyed **1** Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade) – otter recovery restored the balance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Over-fishing of sandeels reduced puffins **1** Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased – over-fishing of sardines / herring (competitors) also increased jellyfish **1** Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction – loss of elephants / large herbivores reduces seed dispersal of trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1 They control abiotic features:** beaver dams create ponds colonised by aquatic species – elephants create clearings and keep water holes open **1** Species that depend on these features (light, water, minerals) cannot survive in their absence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores **1** So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Competitor:** grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites – rhododendron shades ground and releases toxins preventing native regrowth **2 Predator:** American mink (escaped fur farms) reduced water vole range – cane toads in Australia – cats / rats / pigs on islands eating ground-nesting birds – Nile perch in Lake Victoria ate cichlids to extinction **2 Pathogen:** squirrel pox virus carried by grey squirrels kills red squirrels – crayfish plague carried by signal crayfish kills white-clawed crayfish – Dutch elm disease, ash dieback, sudden oak death imported on plants / soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids **1** The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

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